



FACULTY OF APPLIED & COMPUTER SCIENCES

DEPARTMENT OF COMPUTER SCIENCES

DIPLOMA: INFORMATION TEHNOLOGY

SUBJECT : DEVELOPMENT SOFTWARE 3.1
SUBJECT CODE : ASDSX3A
FORMATIVE ASSESSMENT :
DUE DATE : 20 MARCH 2024
DURATION : 4 HOURS
EXAMINER(S) : MRS S MOYO
MODERATOR(S) : MR N SOGANILE

REQUIREMENTS : NOT APPLICABLE

INSTRUCTIONS:

1. Answer all questions.

MARKS: **Total** **= 100**
 Full marks **= 100**

THE QUESTION PAPER CONISTS OF: 4 typed pages

QUESTION 1

[20 Marks]

A class `PizzaPrice` stores the **name** of a pizza and 3 prices, the price of a small pizza(**smallPPrice**), the price of a medium sized pizza (**mediumPPrice**) and the price of a large pizza(**largePPrice**). The class has a constructor that passes as parameters the name and the price of a small pizza only.

The price of a **medium pizza is 60%** more than the price of a small pizza and the price of a **large pizza is 90%** more than the price of a small pizza.

You need to code the `PizzaPrice` class in such a way that when you run the `PizzaPriceTest` class, ALL the test methods in the `PizzaPriceTest` class will pass. The `PizzaPriceTest` class is found is provided. NO changes should be made in the `PizzaPriceTest` class.

- Code the parametrized constructor of the `PizzaPrice` class.
- Code all the mutator methods to only accept/assign a price when it is greater than zero.
- Code the **`setMediumPPrice`** and the **`setLargePPrice`** methods to calculate and assign the price of a medium and large pizza accordingly.

The `setMediumPPrice` and `setLargePPrice` must **round off the price to 2 decimal places before assigning it to the instance variables** (use the **`Math.round()`** method to round off the price.

QUESTION 2

You need to write an application to manage information of products that are sold per item (item- based products) as well as products that are sold by weight (weight-based products). Each product has a barcode of 8 digits. Products that are sold per item have a name, manufacturer, and unit price (in cents). Products that are sold by weight have a name, unit price (in cents per kilogram) as well as a weight (in grams).

Each type of product also needs a method (**calculateTotalCost ()**) to calculate its total cost. For item- based products the total cost is the unit price plus tax. The total cost of a weight-based product is the product's weight multiplied with its unit price plus tax. Use 15% as the tax rate for both types of products.

Example data:

Item- based products			
Barcode	Name of product	Manufacturer	Unit price
178030640224	Milk, 1L	Clover	1550
167832345047	Tuna	I& J	1500

weight- based products			
Barcode	Name of product	Unit price	weight
245134867531	Bananas	4300	540
267897542561	Rump steak	12500	250

Do the following:

Task 1

[31 Marks]

From the given information, identify all necessary classes with their attributes, constructors, and methods. Two concrete classes (**ItemBasedProduct** and **WeightBasedProduct**) and one abstract class **Product** should be among the identified classes. Produce code of all the identified classes. Abstract methods must be used to ensure the re-use of as much code as possible and to ensure polymorphic behaviour. Therefore, no code or instance variables may be duplicated. Do not use instance variables that are dependent on the values of other instance variables. DO NOT code the toString() method for the classes.

The GUI for the application should as shown in Figure 1

Figure 1

Code the application as follows:

Task 2

[5 Marks]

In the `radItemBasedProductActionPerformed(java.awt.event.ActionEvent evt)` write code to:

- Create an item- based product object.
- Add the created object to the array list.
- Clear the controls.

Task 3

[5 Marks]

In the `radWeightBasedProductActionPerformed(java.awt.event.ActionEvent evt)` write code to:

- Create a weight- based product object.
- Add the created object to the array list.
- Clear the controls.

Task 4

[8 Marks]

In the `btnViewSalesActionPerformed(java.awt.event.ActionEvent evt)` write code to:

- Display the data from the array list as shown in Figure 2. The products should be displayed in the order in which they were stored in the array list. **Hint:** make use of **instanceof** operator to check if the object belongs to the `WeightBasedProduct` or `ItemBasedProduct` type.

```

List of products

Barcode 267897542561 is Rump steak, unit price 12500c/Kg, weight 250 grams
Barcode 245134342474 is Bananas, unit price 4300c/Kg, weight 450 grams
Barcode 178030640224 is Milk 1L from Clover, unit price 1550c
Barcode 167832345047 is Tuna from I & J, unit price 1500c
Barcode 251336424031 is Apples, unit price 3350c/Kg, weight 135 grams
Barcode 145672345754 is Oros Orange Squash from Brookes, unit price 3400c
Barcode 245134867531 is Bananas, unit price 4300c/Kg, weight 540 grams
  
```

Figure 2

Task 5

[7 Marks]

In the `btnViewTotalCostActionPerformed(java.awt.event.ActionEvent evt)`, write code to:

- Display from the array list, the barcode number, name and total cost of the product as shown in Figure 3. Hint: use the **`String.format()`** method to format the data as shown.

```
Total cost of each item
Barcode: 267897542561: Rump steak: R35.94
Barcode: 245134342474: Bananas: R22.25
Barcode: 178030640224: Milk 1L: R17.83
Barcode: 167832345047: Tuna: R17.25
Barcode: 251336424031: Apples: R5.20
Barcode: 145672345754: Oros Orange Squash: R39.10
Barcode: 245134867531: Bananas: R26.70
```

Figure 3

Task 6

[4 Marks]

In the `btnViewItemBasedActionPerformed(java.awt.event.ActionEvent evt)`, write code to:

- Display from the array list, item-based products only as shown in **Figure 4**

```
List of item based products Sold

Barcode 178030640224 is Milk 1L from Clover, unit price 1550c
Barcode 167832345047 is Tuna from I & J, unit price 1500c
Barcode 145672345754 is Oros Orange Squash from Brookes, unit price 3400c
```

Figure 4

Task 7

[4 Marks]

In the `btnViewWeightBasedActionPerformed(java.awt.event.ActionEvent evt)`, write code to:

- Display from the array list, weight-based products only as shown in **Figure 5**

```
List of weight based products sold

Barcode 267897542561 is Rump steak, unit price 12500c/Kg, weight 250 grams
Barcode 245134342474 is Bananas, unit price 4300c/Kg, weight 450 grams
Barcode 251336424031 is Apples, unit price 3350c/Kg, weight 135 grams
Barcode 245134867531 is Bananas, unit price 4300c/Kg, weight 540 grams
```

Figure 5

Task 8

[6 Marks]

In the `btnDisplayTotalCostsActionPerformed(java.awt.event.ActionEvent evt)` write code to:

- Calculate and display on a message box, the total cost of weight-based products, item-based products, and the grand total of all the products in the array list as shown in **Figure 6**.

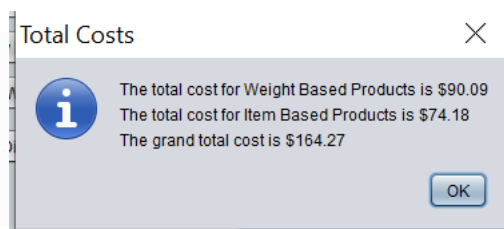


Figure 6

*****The End*****Good Luck*****